src/app/graphicsPerformance.ts

Kind: added · Baseline: 9aadd5eb5604 · Target: NovaDiff · Generated: 2026-05-26T19:32:31.029Z

Overview

A new file `src/app/graphicsPerformance.ts` (lines R1-R16) introduces a single quality profile for docked + expanded code-city rendering. It defines a threshold constant and two helper functions that adjust pixel density and decide when to omit decorative window meshes.

Key changes

- **Exported constant** `CITY_OMIT_WINDOWS_BUILDINGS = 900` (R4) – the building count above which window details are suppressed.
- **Function** `cityPixelRatio(buildingCount: number): number` (R6-R12) –
  - Uses `window.devicePixelRatio` if available, otherwise defaults to 1.
  - Caps the ratio at 1.5 for ≥2000 buildings, otherwise at 2.
- **Function** `shouldOmitWindowDetail(buildingCount: number): boolean` (R14-R16) – returns `buildingCount >= CITY_OMIT_WINDOWS_BUILDINGS`.
- **Documentation comments** (R1-R3) explain the module's purpose and threshold logic.

Impact

- **Performance**: Capping pixel density and omitting window meshes above a configurable threshold reduces rendering load for large cities.
- **Maintainability**: Centralizes graphics tuning in one file, simplifying future adjustments.
- **Compatibility**: The `window` guard ensures the module works in non-browser environments (SSR, tests).

Risks & follow-ups

- **Threshold tuning**: The hard-coded `900` may not suit all use cases; its appropriateness for real-world city sizes is unknown from the diff.
- **Device pixel ratio fallback**: In environments without `window`, the fallback of `1` may under-render; tests should cover this path.
- **Integration**: Verify that the new functions are imported and used by the rendering pipeline; otherwise the file remains unused.
- **Build lint**: Run `npm run lint`, `npm test`, and `npm run build` to catch any TypeScript or import errors introduced by the new module.